

Load Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q23-2347

Chatom 345 kV
September 18, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2347. The Interconnection Customer proposes a 272.7 MW large load (data center) facility interconnecting at the Chatom 345 kV substation in the Virginia area. The request was received on September 07, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 353 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 21 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$87,177,000.

3. Network Impacts

3.1 Load Deliverability Analysis

The CARD – YUCAIPA 138 kV is overloaded to 132.1% of its 267 MVA emergency rating for the loss of the CRANDALL – ASPEN 345 kV L. The project contributes 10.8% DFAX (29.4 MW) to this overload.

The CARD – CRAGGY SUBSTATION 138 kV is overloaded to 129.9% of its 258 MVA emergency rating for the loss of the CRANDALL – ASPEN 345 kV L. The project contributes 10.8% DFAX (29.4 MW) to this overload.

STUARTS DRAFT – CAGGY 138 kV Line: 123.9% loading (239 MVA) following the outage of the ASPEN – CRANDALL 345 kV L; DFAX = 10.8%.

For the loss of CRANDALL – ASPEN 345 kV, the HARRY ALLEN – STUARTS DRAFT 138 kV L loads to 120.3% of its 233 MVA rating. Project DFAX: 10.8%.

For the loss of CRANDALL – DAYTON 345 kV, the REDLAND – VERONA 138 kV line loads to 117.8% of its 298 MVA rating. Project DFAX: 6.9%.

VERONA – GREENBANK 138 kV: 117.6% loading (288 MVA) following the outage of the DAYTON – CRANDALL 345 kV; DFAX = 6.9%.

The DECLO – SALAMANCAROCHESTER ST 138 kV is overloaded to 114.3% of its 347 MVA emergency rating for the loss of the ASPEN SUBSTATION – CRANDALL 345 kV. The project contributes 12.5% DFAX (34.1 MW) to this overload.

The AUSTIN – DECLO 138 kV #2 is overloaded to 112.7% of its 332 MVA emergency rating for the loss of the LOWMOOR – ASPEN 345 kV line. The project contributes 5.3% DFAX (14.6 MW) to this overload.

For the loss of ASPEN – LOWMOOR 345 kV, the DECLO – AUSTIN 138 kV Line 3 loads to 112.7% of its 332 MVA rating. Project DFAX: 5.3%.

The AUSTIN – DECLO 138 kV is overloaded to 112.6% of its 332 MVA emergency rating for the loss of the ASPEN – LOWMOOR 345 kV. The project contributes 5.3% DFAX (14.6 MW) to this overload.

The AUSTIN – BATH COUNTY 138 kV Cir. 1 is overloaded to 111.6% of its 338 MVA emergency rating for the loss of the ASPEN – LOWMOOR 345 kV. The project contributes 5.3% DFAX (14.6 MW) to this overload.

For the loss of ASPEN – LOWMOOR SUBSTATION 345 kV line, the AUSTIN – BATH COUNTY 138 kV Line 3 loads to 111.4% of its 339 MVA rating. Project DFAX: 5.3%.

For the loss of ASPEN – CRANDALL 345 kV, the FISHERSVILLE – HARRY ALLEN 138 kV loads to 110.5% of its 219 MVA rating. Project DFAX: 10.8%.

For the loss of CRANDALL – ASPEN 345 kV, the ERIE – FISHERSVILLE 138 kV line loads to 107.9% of its 201 MVA rating. Project DFAX: 10.8%.

The LOWMOOR 345/138 kV TX is overloaded to 106.9% of its 723 MVA emergency rating for the loss of the ASPEN – LOWMOOR 345 kV. The project contributes 5.4% DFAX (14.8 MW) to this overload.

For the loss of LOWMOOR – ASPEN 345 kV, the FOLSOM – ST BARNABAS 138 kV line loads to 104.7% of its 255 MVA rating. Project DFAX: 9.7%.

The OAK CRK STATION SUBSTATION – SANS SOUCI SUBSTATION 138 kV is overloaded to 103.9% of its 372 MVA emergency rating for the loss of the DAYTON – CHATOM 345 kV. The project contributes 21.4% DFAX (58.4 MW) to this overload.

The HARRISONBURG – SANS SOUCI 138 kV line is overloaded to 103.9% of its 372 MVA emergency rating for the loss of the DAYTON – CHATOM 345 kV. The project contributes 21.4% DFAX (58.4 MW) to this overload.

BATH COUNTY – ASTIN 138 kV: 103.5% loading (364 MVA) following the outage of the ASPEN SUBSTATION – LOWMOOR 345 kV; DFAX = 5.3%.

For the loss of ASPEN – CRANDALL 345 kV line, the SALAMANCAROCHESTER ST – REDLAND 138 kV loads to 102.6% of its 234 MVA rating. Project DFAX: 11.6%.

OAK CREEK STATION – WADSWORTH 138 kV: 101.1% loading (328 MVA) following the outage of the CHATOM – DAYTON 345 kV; DFAX = 21.4%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
CARD – YUCAIPA 138 kV	Rebuild line with higher-capacity conductor	\$4,381,000	37% (\$1,633,000)
CARD – CRAGGY 138 kV line	Rebuild line with higher-capacity conductor	\$9,419,000	37% (\$3,511,000)
CRAGGY – STUARTS DRAFT 138 kV L	Rebuild line with higher-capacity conductor	\$4,424,000	37% (\$1,649,000)
HARRY ALLEN – STUARTS DRAFT 138 kV L	Rebuild line with higher-capacity conductor	\$6,772,000	37% (\$2,524,000)
REDLAND – VERONA 138 kV line	Rebuild line with higher-capacity conductor	\$5,670,000	34% (\$1,914,000)
VERONA – GREENBANK 138 kV	Rebuild line with higher-capacity conductor	\$3,662,000	34% (\$1,236,000)
SALAMANCAROCHESTER ST – DECLO 138 kV	Rebuild line with higher-capacity conductor	\$41,118,000	35% (\$14,289,000)
AUSTIN – DECLO 138 kV #2	Rebuild line with higher-capacity conductor	\$14,757,000	59% (\$8,727,000)
AUTIN – DECLO 138 kV Ckt 3	Rebuild line with higher-capacity conductor	\$12,898,000	59% (\$7,627,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
DECLO – AUSTIN 138 kV	Rebuild line with higher-capacity conductor	\$18,968,000	59% (\$11,217,000)
AUSTIN – BATH COUNTY 138 kV Cir. 1	Rebuild line with higher-capacity conductor	\$15,429,000	59% (\$9,124,000)
AUSTIN – BATH COUNTY 138 kV Line 3	Rebuild line with higher-capacity conductor	\$25,429,000	59% (\$15,037,000)
FISHERSVILLE – HARRY ALLEN 138 kV	Rebuild line with higher-capacity conductor	\$2,774,000	37% (\$1,034,000)
ERIE – FISHERSVILLE 138 kV	Reconductor line	\$1,195,000	37% (\$445,000)
LOWMOOR 345/138 kV TX	Replace transformer with higher-rated unit	\$15,766,000	3% (\$430,000)
FOLSOM – ST BARNABAS 138 kV	Reconductor line	\$3,379,000	28% (\$942,000)
OAK CRK STATION SUBSTATION – SANS SOUCI SUBSTATION 138 kV	Reconductor line	\$897,000	24% (\$216,000)
HARRISONBURG – SANS SOUCI 138 kV	Reconductor line	\$1,248,000	24% (\$300,000)
BATH COUNTY – ASTIN 138 kV	Reconductor line	\$4,396,000	59% (\$2,600,000)
SALAMANCAROCHESTER ST – REDLAND 138 kV	Reconductor line	\$5,703,000	38% (\$2,175,000)
OAK CREEK STATION – WADSWORTH 138 kV	Reconductor line	\$2,272,000	24% (\$547,000)
	Total allocated to project		\$87,177,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.